

Chapter 1

Optimization

1.1 Stochastic optimization

We consider optimization of stochastic objectives of the form

$$\mathcal{L}(\boldsymbol{\theta}) = \mathbb{E}_{q_{\boldsymbol{\theta}}(\mathbf{z})} [\tilde{L}(\boldsymbol{\theta}, \mathbf{z})]$$

where $\mathbf{z}$ is a r.v., such as an external noise

1.1.1 SGD for optimizing the parameters of a distribution

$\mathbf{z}$ could be a latent variable sampled from an inference network $q_{\boldsymbol{\theta}}$.

$$\begin{aligned} \nabla_{\boldsymbol{\theta}} \mathbb{E}_{q_{\boldsymbol{\theta}}(\mathbf{z})} [\tilde{L}(\boldsymbol{\theta}, \mathbf{z})] &= \nabla_{\boldsymbol{\theta}} \int \mathcal{L}(\boldsymbol{\theta}, \mathbf{z}) q_{\boldsymbol{\theta}}(\mathbf{z}) d\mathbf{z} = \int \nabla_{\boldsymbol{\theta}} \tilde{L}(\boldsymbol{\theta}, \mathbf{z}) q_{\boldsymbol{\theta}}(\mathbf{z}) d\mathbf{z} \\ &= \int [\nabla_{\boldsymbol{\theta}} \tilde{L}(\boldsymbol{\theta}, \mathbf{z})] q_{\boldsymbol{\theta}}(\mathbf{z}) d\mathbf{z} + \int \tilde{L}(\boldsymbol{\theta}, \mathbf{z}) [\nabla_{\boldsymbol{\theta}} q_{\boldsymbol{\theta}}(\mathbf{z})] d\mathbf{z} \end{aligned}$$

The first term can be approximated by

$$\int [\nabla_{\boldsymbol{\theta}} \tilde{L}(\boldsymbol{\theta}, \mathbf{z})] q_{\boldsymbol{\theta}}(\mathbf{z}) d\mathbf{z} \approx \frac{1}{S} \sum_{s=1}^S \nabla_{\boldsymbol{\theta}} \tilde{L}(\boldsymbol{\theta}, \mathbf{z}_s).$$

where $\mathbf{z}_s \sim q_{\boldsymbol{\theta}}$.

Consider the second term,

$$I \stackrel{\text{def}}{=} \int \tilde{L}(\boldsymbol{\theta}, \mathbf{z}) [\nabla_{\boldsymbol{\theta}} q_{\boldsymbol{\theta}}(\mathbf{z})] d\mathbf{z}$$

There are two main methods to approximate I .

1.1.2 Score function estimator (REINFORCE)

$$\nabla_{\boldsymbol{\theta}} q_{\boldsymbol{\theta}}(\mathbf{z}) = q_{\boldsymbol{\theta}}(\mathbf{z}) \nabla_{\boldsymbol{\theta}} \log q_{\boldsymbol{\theta}}(\mathbf{z})$$

So

$$I = \mathbb{E}_{q_{\boldsymbol{\theta}}(\mathbf{z})} [\tilde{L}(\boldsymbol{\theta}, \mathbf{z}) \nabla_{\boldsymbol{\theta}} \log q_{\boldsymbol{\theta}}(\mathbf{z})].$$

which can be approximated by Monte Carlo.

1.1.3 Reparameterization trick

Score function estimator can have *high variance*.

Assumption 1.1.1

- $\tilde{\mathcal{L}}$ is differentiable wrt z
- We can compute a sample from $q_{\theta}(z)$ by first sampling ϵ from some noise distribution q_0 which is independent of θ , and then transforming z using a deterministic and differentiable function $z = g(\theta, \epsilon)$.

$$\mathcal{L}(\theta) = \mathbb{E}_{q_0(\epsilon)} [\tilde{\mathcal{L}}(\theta, g(\theta, \epsilon))]$$

$$\nabla_{\theta} \mathcal{L}(\theta) = \mathbb{E}_{q_0(\epsilon)} [\nabla_{\theta} \tilde{\mathcal{L}}(\theta, g(\theta, \epsilon))] \approx \frac{1}{S} \sum_{s=1}^S \nabla_{\theta} \tilde{\mathcal{L}}(\theta, g(\theta, \epsilon_s)).$$

where $\epsilon_s \sim q_0$.

A example

Let $z \sim \mathcal{N}(\mu, v)$, $v = \sigma^2$.

$$\epsilon \sim \mathcal{N}(0, 1), \quad z = \mu + \sqrt{v}\epsilon$$

$$\mathcal{L}(\mu, v) = \mathbb{E}_{z \sim \mathcal{N}(\mu, v)} [\tilde{\mathcal{L}}(z)] = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, 1)} [\tilde{\mathcal{L}}(\mu + \sqrt{v}\epsilon)]$$

Now the stochasticity is independent of μ and v , so we can compute the gradient by backpropagation. Let

$$z(\mu, v, \epsilon) = \mu + \sqrt{v}\epsilon,$$

$$\frac{\partial \mathcal{L}}{\partial \mu} = \frac{\partial}{\partial \mu} \mathbb{E}_{\epsilon} [\tilde{\mathcal{L}}(z)] = \mathbb{E}_{\epsilon} \left[\frac{\partial \tilde{\mathcal{L}}}{\partial z} \cdot \frac{\partial z}{\partial \mu} \right] = \mathbb{E}_{\epsilon} \left[\frac{\partial \tilde{\mathcal{L}}(z)}{\partial z} \right]$$

$$\frac{\partial \mathcal{L}}{\partial v} = \mathbb{E}_{\epsilon} \left[\frac{\partial \tilde{\mathcal{L}}}{\partial z} \cdot \frac{\partial z}{\partial v} \right] = \mathbb{E}_{\epsilon} \left[\frac{\partial \tilde{\mathcal{L}}(z)}{\partial z} \cdot \frac{\epsilon}{2\sqrt{v}} \right]$$

Then we can use Monte Carlo to compute this.